

Math 250 Week 2

DZB

August 27, 2020

Negation problem. Negate the following:

$\forall \epsilon > 0, \exists \delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$

$\neg(\forall \epsilon > 0, \exists \delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon)$

$\forall \epsilon > 0, \exists \delta > 0$ such that if P , then Q

$\neg(\forall x, P(x)) = \exists x$ such that $\neg P(x)$

$\exists \epsilon > 0$ such that $\neg(\exists \delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon)$

$\exists \epsilon > 0$ such that $\forall \delta > 0, \neg$ (if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon)$

$\neg(P \Rightarrow Q) = P \wedge \neg Q$

$\exists \epsilon > 0$ such that $\forall \delta > 0, |x - x_0| < \delta$ and $|f(x) - f(x_0)| \geq \epsilon$

Problem. Let n be an integer. Prove that $n(n + 1)$ is even.

(This is the same problem as If n is an integer, then $n(n + 1)$ is even.)

Proof (by cases). Suppose that n is an integer. We will break the proof into 2 cases. In the first case, we assume that n is even. In this case, there exists an integer m such that $n = 2m$. Thus, $n(n + 1) = 2m(n + 1)$. This is twice an integer, therefore $n(n + 1)$ is even.

In the second case, n is odd. Thus there exists an integer m such that $n = 2m + 1$. Then, then quantity $n(n + 1)$ is equal to $(2m + 1)(2m + 1 + 1) = (2m + 1)(2m + 2) = 2(2m + 1)(m + 1)$. We thus have an equation $n(n + 1) = 2c$, where $c = (2m + 1)(m + 1)$. Since c is an integer, we conclude that $n(n + 1)$ is even. (QED)

Problem. Prove that for all integers a and b , $ab(a - b)$ is even.

Proof (by cases). Assume that $a, b \in \mathbf{Z}$. We consider the following 3 cases: a is even, b is even, or both are odd. Case 1: assume that a is even. Then there exists $m \in \mathbf{Z}$ such that $a = 2m$. Then $ab(a - b) = 2mb(a - b) = 2(mb(a - b))$. Since $mb(a - b)$ is an integer, we conclude that $ab(a - b)$ is even. Case 2 is identical to case 1.

Case 3: assume that a and b are both odd. Then there exists integers $m, n \in \mathbf{Z}$ such that $a = 2m + 1$ and $b = 2n + 1$. Then $ab(a - b) = ab(2m + 1 - (2n + 1)) = ab(2)(m - n)$ which is twice an integer. We conclude that $ab(a - b)$ is even. QED

Problem. Suppose that $n \in \mathbf{Z}_{\geq 0}$. Then $3 \mid 4^n - 1$.